

Race Day Management System – API Endpoint Specification

Overview

This document outlines all API endpoints for the Race Day Management System organised by functional group. Each endpoint specifies the HTTP method, path, request/response structure, and status codes.

Area	Operation	HTTP Method	Endpoint Path	Request Body	Response (Success)	Status Code
Auth	User Registration	POST	/auth/users	{ "username": "...", "email": "...", "password": "...", "role": "..." }	{ "user_id": "...", "username": "...", "email": "...", "role": "...", "created_at": "..." }	201
	User Login	POST	/auth/login	{ "username": "...", "password": "..." }	{ "user_id": "...", "username": "...", "email": "...", "role": "...", "token": "..." }	200
	Get User Profile	GET	/profiles/{user_id}	N/A	{ "user_id": "...", "username": "...", "email": "...", "role": "...", "created_at": "..." }	200

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Assignment Part 1

	Update Organiser Profile	PUT	/organisers/{organiser_id}	{ "first_name": "...", "last_name": "...", "phone": "..." }	{ "organiser_id": "...", "first_name": "...", "last_name": "...", "phone": "...", "updated_at": "..." }	200
	Update Participant Profile	PUT	/participants/{participant_id}	{ "first_name": "...", "last_name": "...", "phone": "..." }	{ "participant_id": "...", "first_name": "...", "last_name": "...", "phone": "...", "updated_at": "..." }	200
	Delete user account	DELETE	/users/{user_id}	N/A	{ "message": "User deleted successfully" }	204
Events	Create event	POST	/events	{ "event_name": "...", "location": "...", "latitude": "...", "longitude": "...", "event_date": "...", "status": "..." }	{ "event_id": "...", "event_name": "...", "location": "...", "latitude": "...", "longitude": "...", "event_date": "...", "status": "...", "created_at": "..." }	201
	List all events	GET	/events	N/A	{ "events": [{ "event_id": "...", "event_name": "...", "location": "...", "event_date": "...", "status": "...", "created_at": "..." }, ...] }	200

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	Get event details	GET	/events/{event_id}	N/A	{ "event_id": "...", "event_name": "...", "location": "...", "latitude": "...", "longitude": "...", "event_date": "...", "status": "...", "created_at": "..." }	200
	Update event	PUT	/events/{event_id}	{ "event_name": "...", "location": "...", "latitude": "...", "longitude": "...", "event_date": "...", "status": "..." }	{ "event_id": "...", "event_name": "...", "location": "...", "latitude": "...", "longitude": "...", "event_date": "...", "status": "...", "updated_at": "..." }	200
	Delete event	DELETE	/events/{event_id}	N/A	{ "message": "Event deleted successfully" }	204
Categories	Create category	POST	/categories	{ "category_name": "..." }	{ "category_id": "...", "category_name": "...", "created_at": "..." }	201

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	List all Categories	GET	/categories	N/A	{ "categories": [{ "category_id": "...", "category_name": "...", "created_at": "..." }, ...] } }	200
	Get Category details	GET	/categories/{category_id}	N/A	{ "category_id": "...", "category_name": "...", "created_at": "..." }	200
	Update Category	PUT	/categories/{category_id}	{ "category_name": "..." }	{ "category_id": "...", "category_name": "...", "updated_at": "..." }	200
	Delete Category	DELETE	/category/{category_id}	N/A	{ "message": "Category deleted successfully" }	204
Enrolments	Enrol Participant in Event	POST	/enrolments	{ "participant_id": "...", "category_id": "...", "entry_date": "...", "status": "..." }	{ "entry_id": "...", "participant_id": "...", "event_id": "...", "category_id": "...", "entry_date": "...", "status": "...", "results_id": "..." }	201

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List Event Enrolments	GET	/enrolments	N/A	{ "enrolments": [{ "entry_id": "...", "participant_id": "...", "category_id": "...", "entry_date": "...", "status": "..." }, ...] } }	200
Get enrolment Details	GET	/enrolments/{enrolment_id}	N/A	{ "entry_id": "...", "participant_id": "...", "event_id": "...", "category_id": "...", "entry_date": "...", "status": "...", "results_id": "..." }	200
Update enrolment status	PUT	/events/{event_id}/enrolments/{enrolment_id}	{ "status": "..." }	{ "entry_id": "...", "participant_id": "...", "event_id": "...", "category_id": "...", "status": "...", "updated_at": "..." }	200
Participant withdraw from Event	DELETE	/events/{event_id}/enrolments/{enrolment_id}	N/A	{ "message": "Enrolment cancelled successfully" }	204

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Routes	Create Route for an Event	POST	/events/{event_id}/routes	{ "route_name": "...", "distance": "...", "start_location": "...", "end_location": "...", "map_url": "..." }	{ "route_id": "...", "event_id": "...", "route_name": "...", "distance": "...", "start_location": "...", "end_location": "...", "map_url": "..." }	201
	List Routes for the Event	GET	/events/{events_id}/routes	N/A	{ "routes": [{ "route_id": "...", "route_name": "...", "distance": "...", "start_location": "...", "end_location": "...", "map_url": "..." }, ...] } }	200
	Get Route Details	GET	/events/{events_id}/routes/{route_id}	N/A	{ "route_id": "...", "event_id": "...", "route_name": "...", "distance": "...", "start_location": "...", "end_location": "...", "map_url": "..." }	200

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	Update route	PUT	/events/{events_id}/routes/{route_id}	{ "route_name": "...", "distance": "...", "start_location": "...", "end_location": "...", "map_url": "..." }	{ "route_id": "...", "event_id": "...", "route_name": "...", "distance": "...", "start_location": "...", "end_location": "...", "map_url": "...", "updated_at": "..." }	200
	Delete Route	DELETE	/events/{event_id}/routes/{route_id}	N/A	{ "message": "Route deleted successfully" }	204
Results	Record Race Results	POST	/events/{event_id}/results	{ "results_id": "...", "participant_id": "...", "start_time": "...", "finish_time": "...", "position": "..." }	{ "results_id": "...", "participant_id": "...", "event_id": "...", "start_time": "...", "finish_time": "...", "position": "..." }	201
	Get Race Results	GET	/results/{results_id}	N/A	{ "results_id": "...", "participant_id": "...", "event_id": "...", "start_time": "...", "finish_time": "...", "position": "..." }	200

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Get Participant Performance History	GET	/results/history/{participant_id}	N/A	{ "history": [{ "results_id": "...", "event_id": "...", "finish_time": "...", "position": "...", "recorded_at": "..." }, ...] } }	200
Update Race Results	PUT	/results/{results_id}	{ "start_time": "...", "finish_time": "...", "position": "..." }	{ "results_id": "...", "participant_id": "...", "event_id": "...", "start_time": "...", "finish_time": "...", "position": "...", "updated_at": "..." }	200
Delete Race Results	DELETE	/results/{results_id}	N/A	{ "message": "Result deleted successfully" }	204

Notes

Timestamps

- All timestamps (created_at, updated_at, recorded_at) should be in ISO 8601 format (e.g., "2026-09-11T14:30:00Z")

Placeholders

- {user_id}, {event_id}, {category_id}, {organiser_id}, {participant_id}, {entry_id}, {route_id}, {results_id} are placeholders for actual IDs in the URL path
- The "..." in request/response bodies represents placeholder values that will be replaced with actual data

Weather Integration

- Weather data is retrieved from an external API using event latitude and longitude coordinates
- No dedicated weather endpoint is needed — the weather API is called on the frontend using coordinates from the Events endpoint

Routes and Maps

- Routes can be displayed using Google Maps integration on the frontend
- The map_url field contains the URL or embed code for the route's map display

Authentication

- Protected endpoints require an authentication token in the request headers
- Header format: Authorization: Bearer {token}
- The token is obtained from the User Login endpoint response